Introduction to Linear Programming

- Linear Programming Problem
- Problem Formulation
- A Simple Maximization Problem
- Graphical Solution Procedure
- Extreme Points and the Optimal Solution
- Computer Solutions
- A Simple Minimization Problem
- Special Cases

Linear Programming

- Linear programming has nothing to do with computer programming.
- The use of the word "programming" here means "choosing a course of action."
- Linear programming involves choosing a course of action when the mathematical model of the problem contains only linear functions.

Linear Programming (LP) Problem

- The maximization or minimization of some quantity is the objective in all linear programming problems.
- All LP problems have constraints that limit the degree to which the objective can be pursued.
- A feasible solution satisfies all the problem's constraints.
- An optimal solution is a feasible solution that results in the largest possible objective function value when maximizing (or smallest when minimizing).
- A graphical solution method can be used to solve a linear program with two variables.

Linear Programming (LP) Problem

- If both the objective function and the constraints are linear, the problem is referred to as a linear programming problem.
- Linear functions are functions in which each variable appears in a separate term raised to the first power and is multiplied by a constant (which could be 0).
- Linear constraints are linear functions that are restricted to be "less than or equal to", "equal to", or "greater than or equal to" a constant.

Problem Formulation

- Problem formulation or modeling is the process of translating a verbal statement of a problem into a mathematical statement.
- Formulating models is an art that can only be mastered with practice and experience.
- Every LP problems has some unique features, but most problems also have common features.
- General guidelines for LP model formulation are illustrated on the slides that follow.

Guidelines for Model Formulation

- Understand the problem thoroughly.
- Describe the objective.
- Describe each constraint.
- Define the decision variables.
- Write the objective in terms of the decision variables.
- Write the constraints in terms of the decision variables.

Example 1: A Simple Maximization Problem

$$\begin{array}{lrcl} \text{Max} & 5x_1 + 7x_2 & & \\ \text{s.t.} & x_1 & \le & 6 \\ & 2x_1 + 3x_2 & \le & 19 \\ & x_1 + x_2 & \le & 8 \end{array}$$

$$x_1 \ge 0 \text{ and } x_2 \ge 0$$

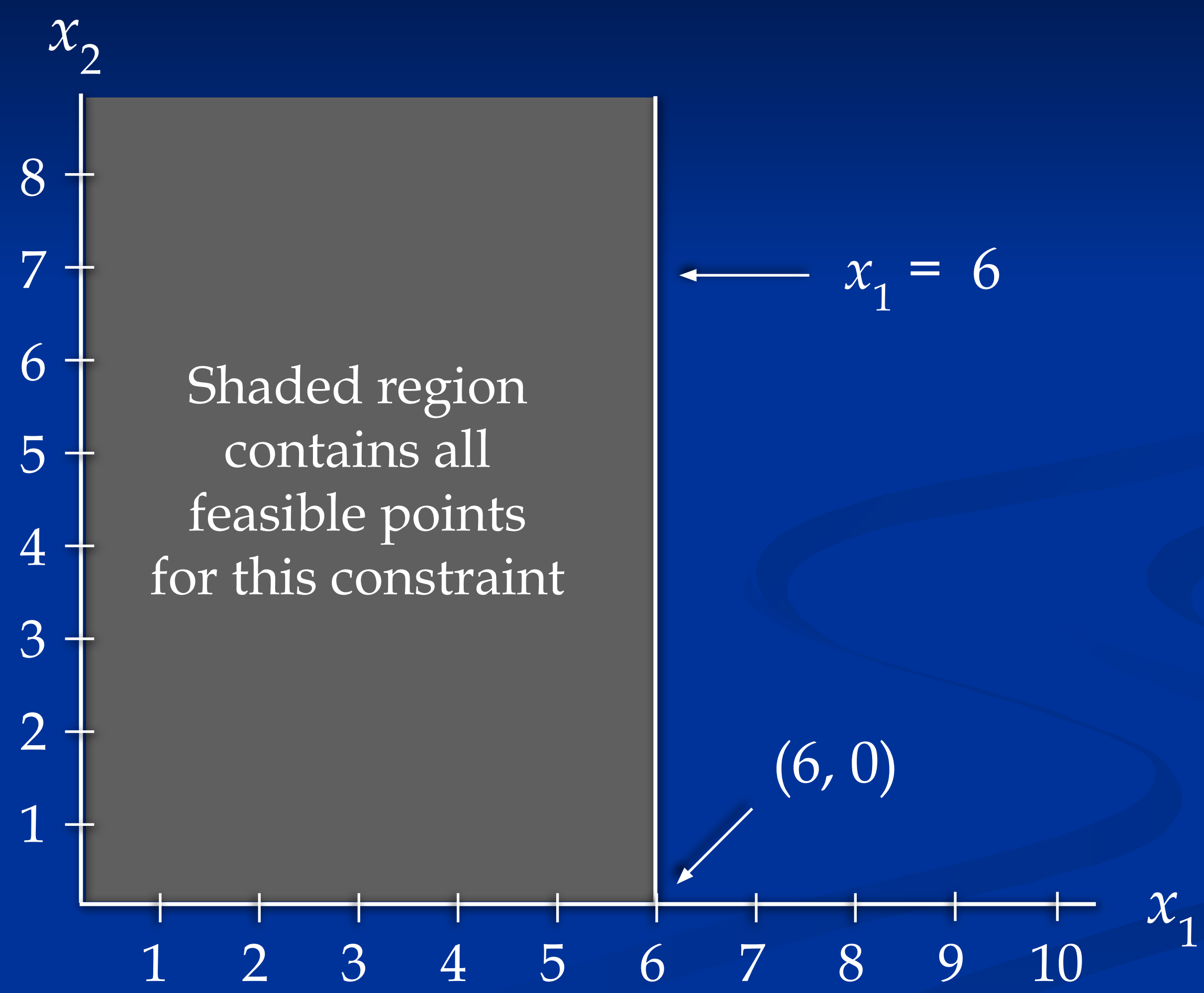
Objective Function

“Regular” Constraints

Non-negativity Constraints

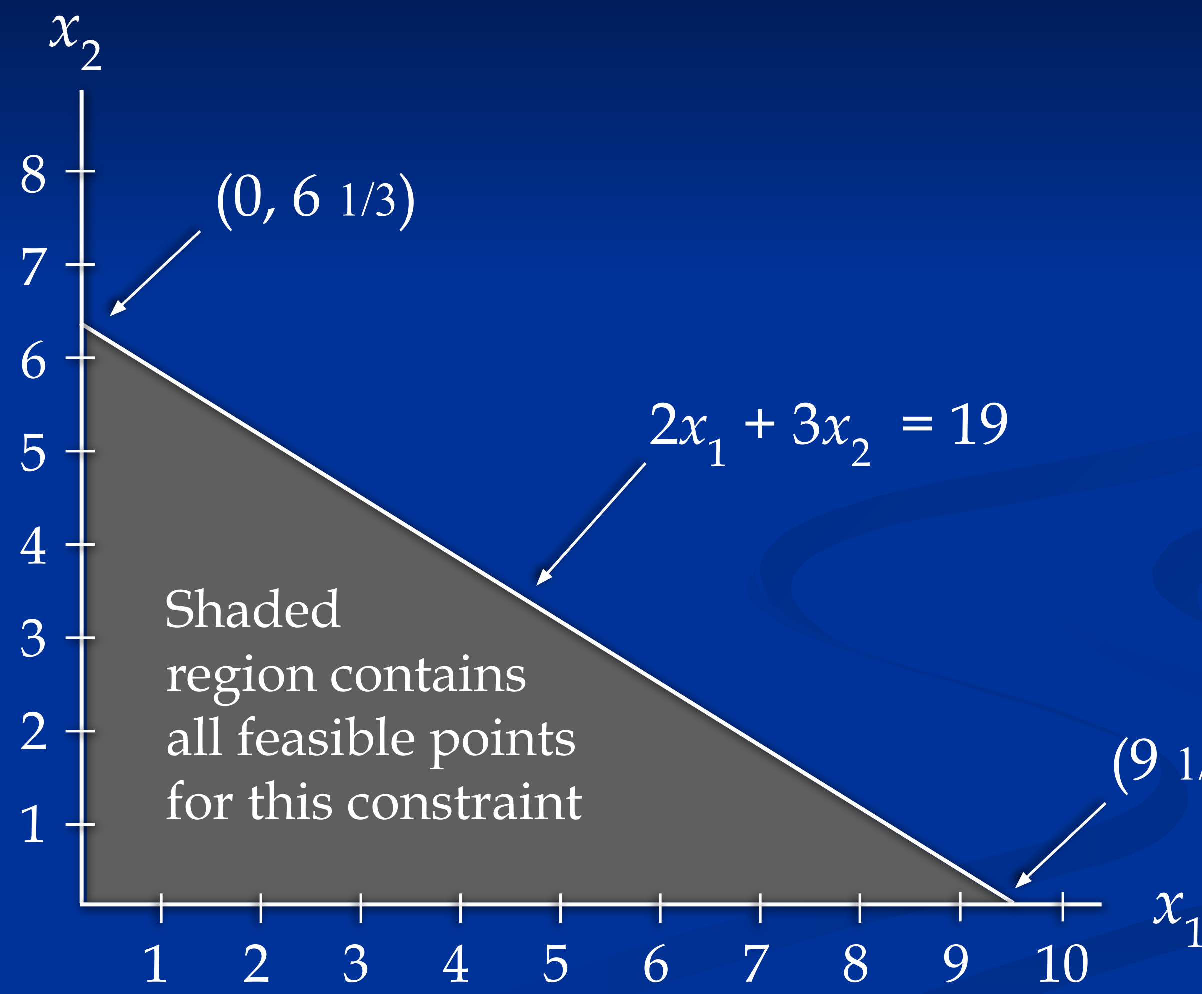
Example 1: Graphical Solution

- First Constraint Graphed



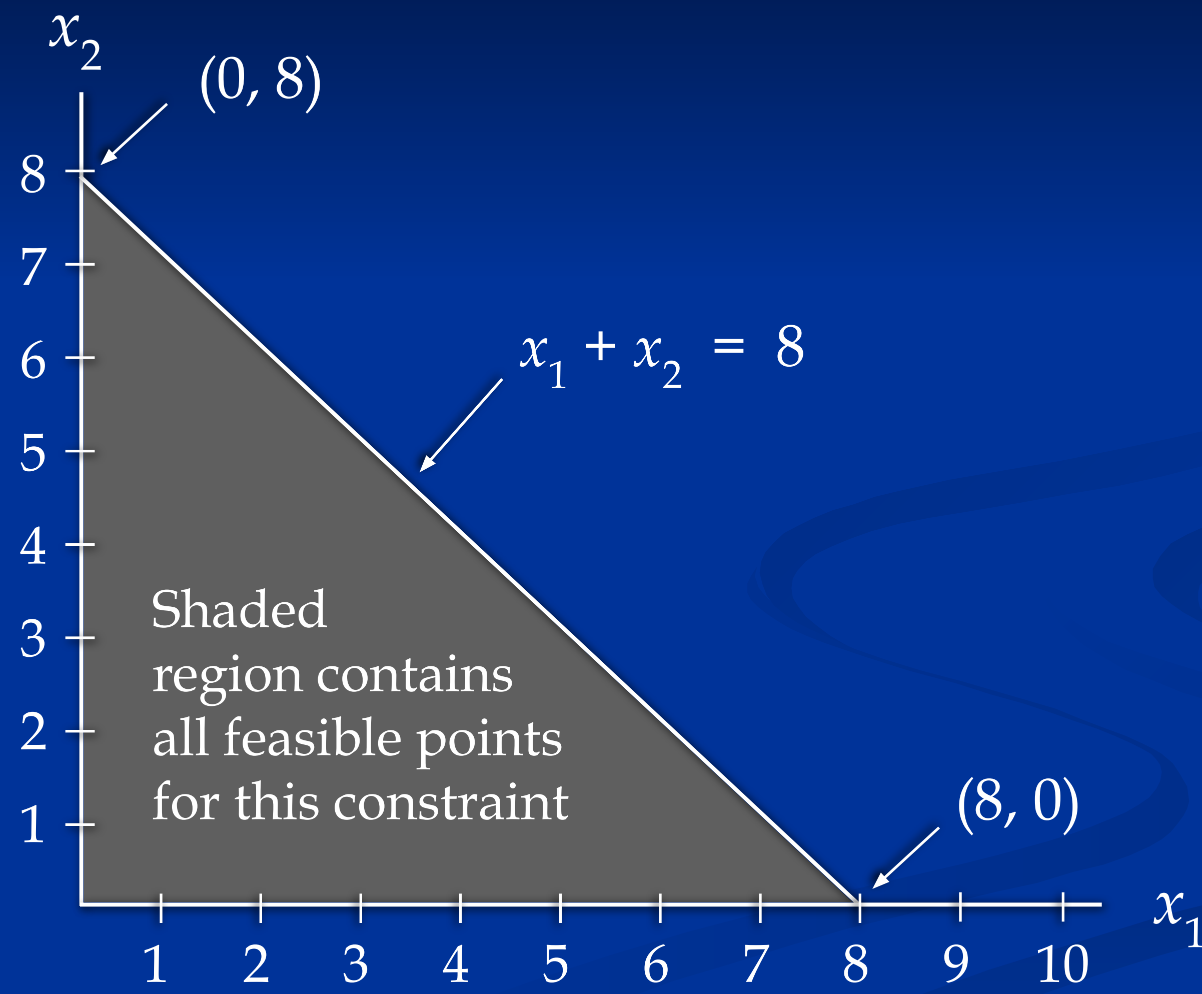
Example 1: Graphical Solution

- Second Constraint Graphed



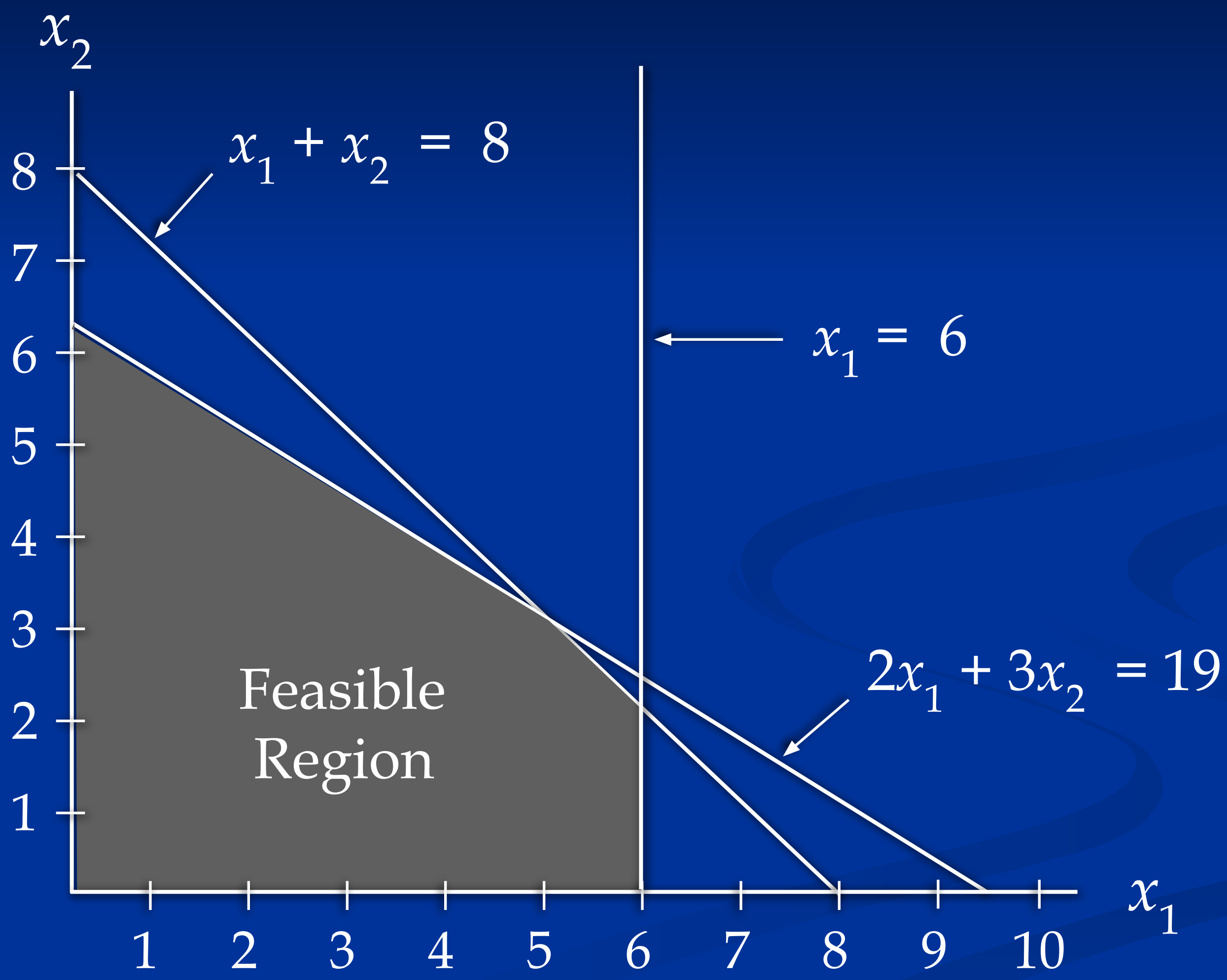
Example 1: Graphical Solution

- Third Constraint Graphed



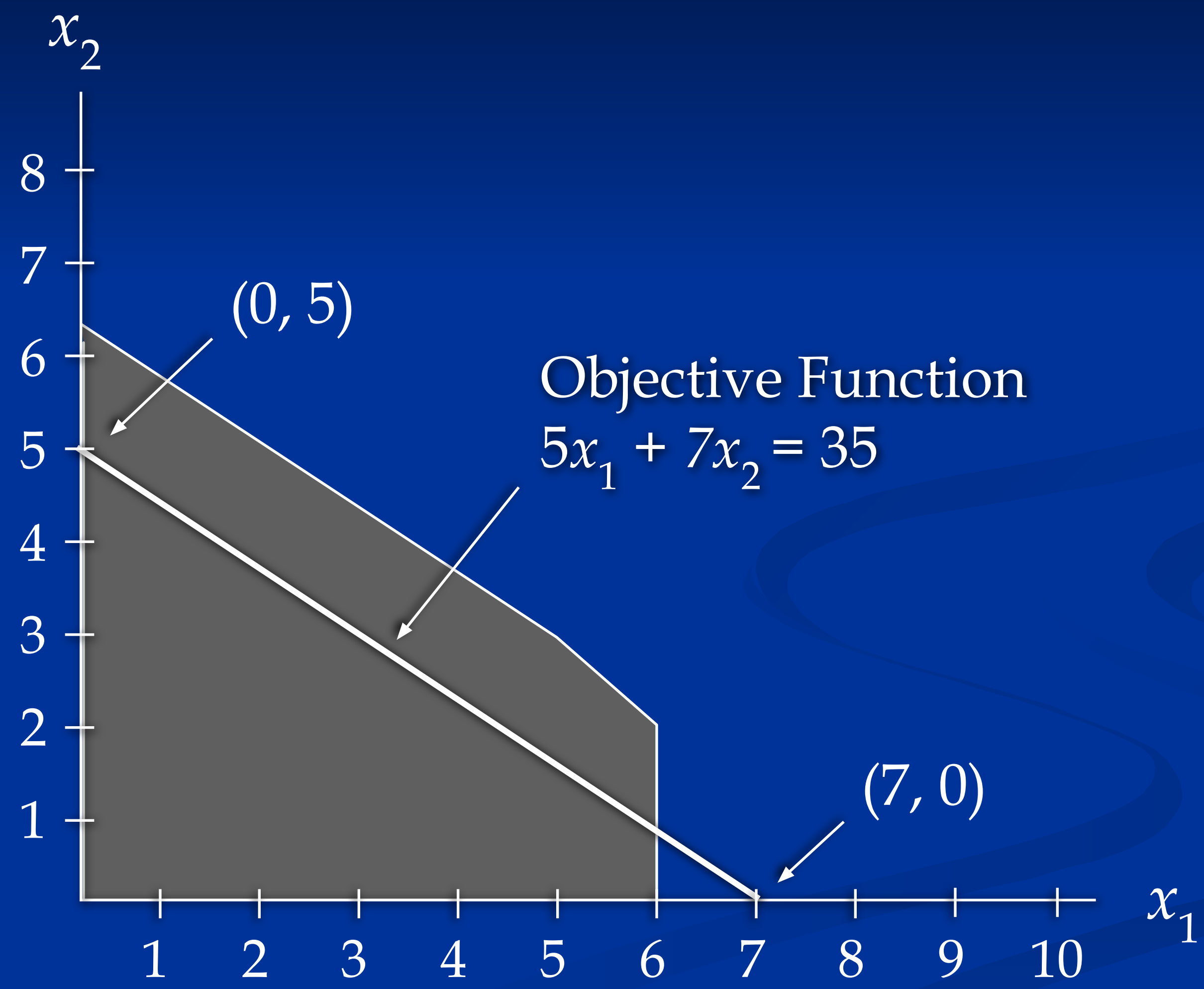
Example 1: Graphical Solution

- Combined-Constraint Graph Showing Feasible Region



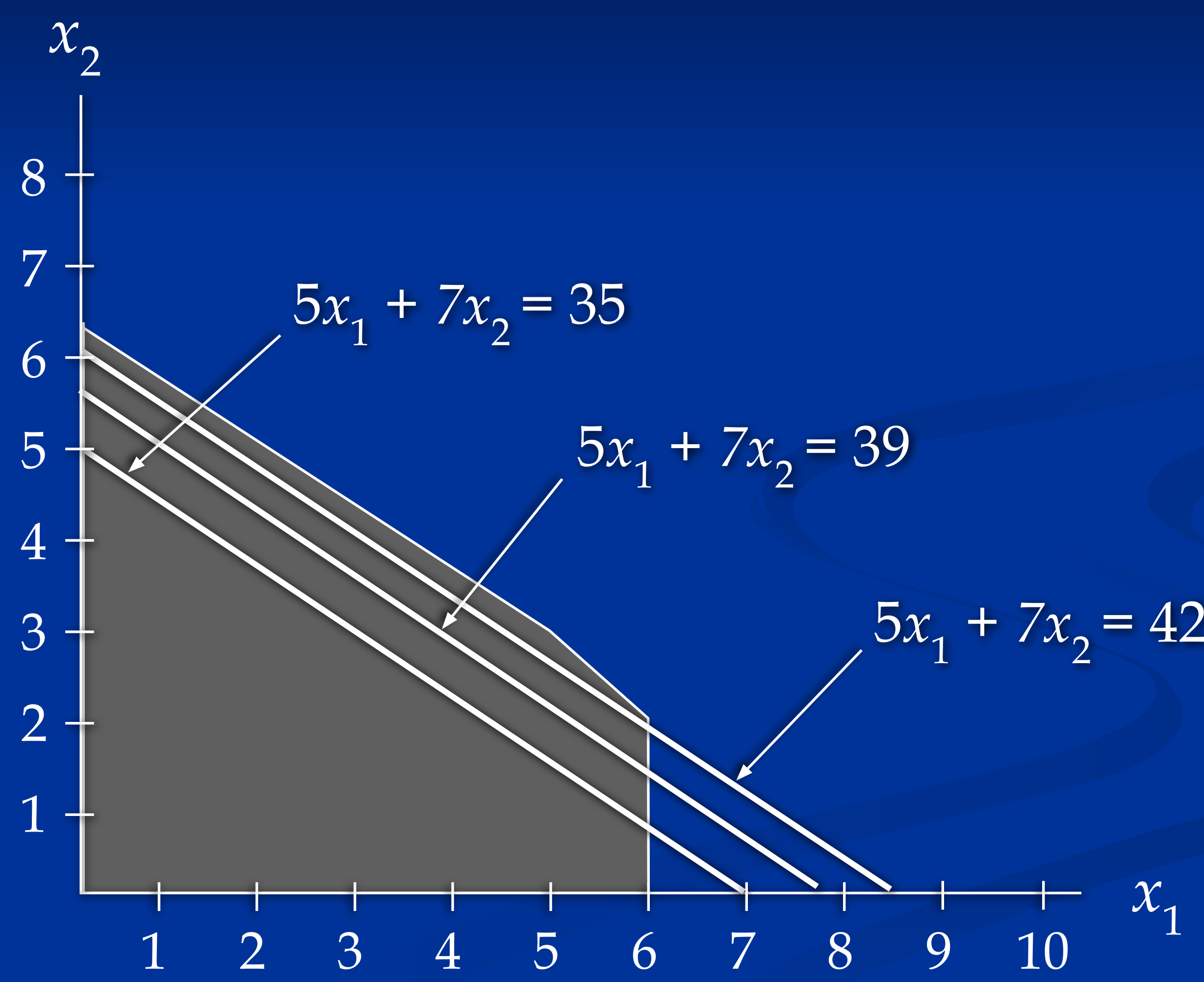
Example 1: Graphical Solution

- Objective Function Line



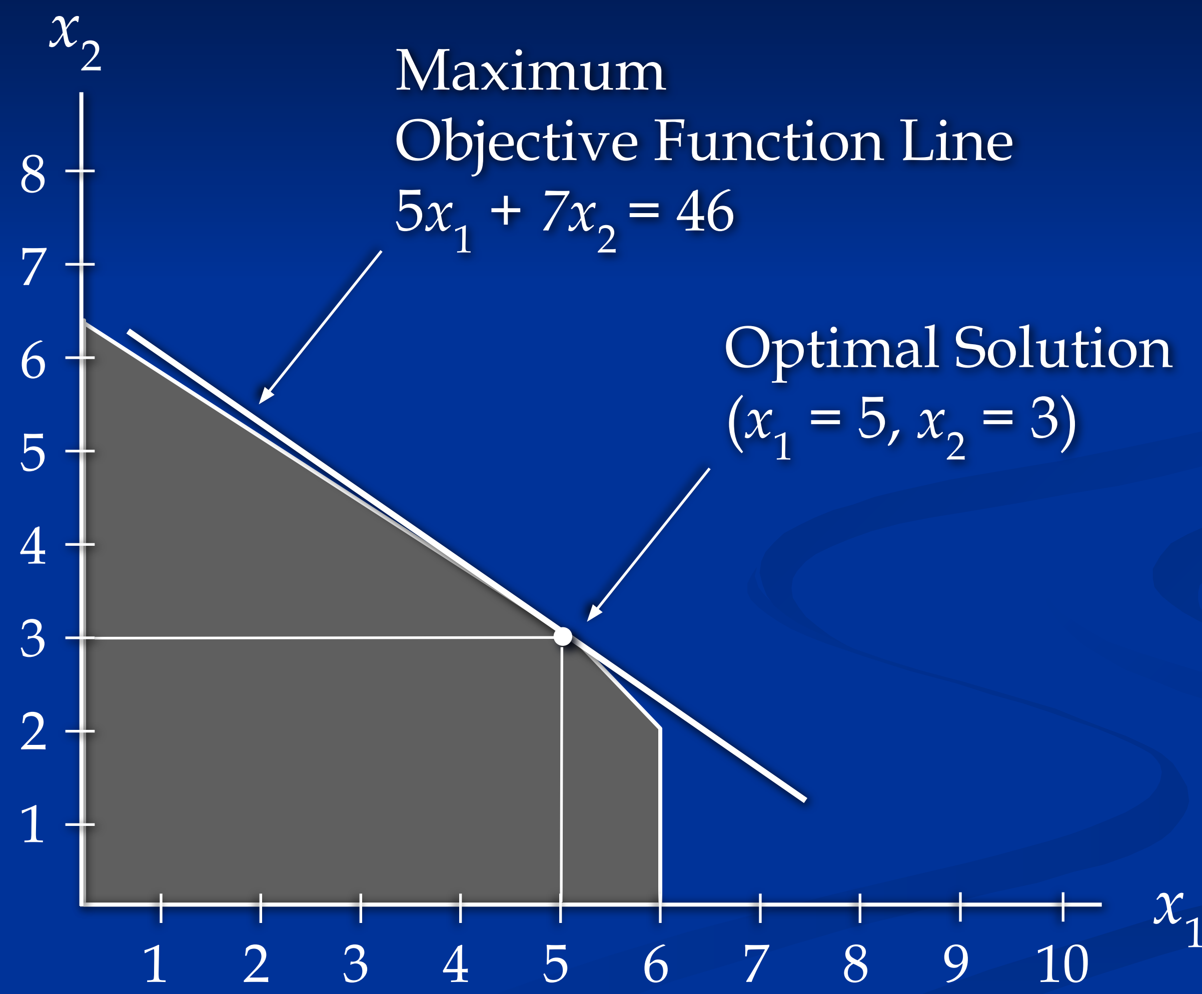
Example 1: Graphical Solution

- Selected Objective Function Lines



Example 1: Graphical Solution

- Optimal Solution



Summary of the Graphical Solution Procedure for Maximization Problems

- Prepare a graph of the feasible solutions for each of the constraints.
- Determine the feasible region that satisfies all the constraints simultaneously.
- Draw an objective function line.
- Move parallel objective function lines toward <u>larger</u> objective function values without entirely leaving the feasible region.
- Any feasible solution on the objective function line with the <u>largest</u> value is an optimal solution.

Slack and Surplus Variables

- A linear program in which all the variables are non-negative and all the constraints are equalities is said to be in standard form.
- Standard form is attained by adding slack variables to "less than or equal to" constraints, and by subtracting surplus variables from "greater than or equal to" constraints.
- Slack and surplus variables represent the difference between the left and right sides of the constraints.
- Slack and surplus variables have objective function coefficients equal to 0.

Slack Variables (for $\leq$ constraints)

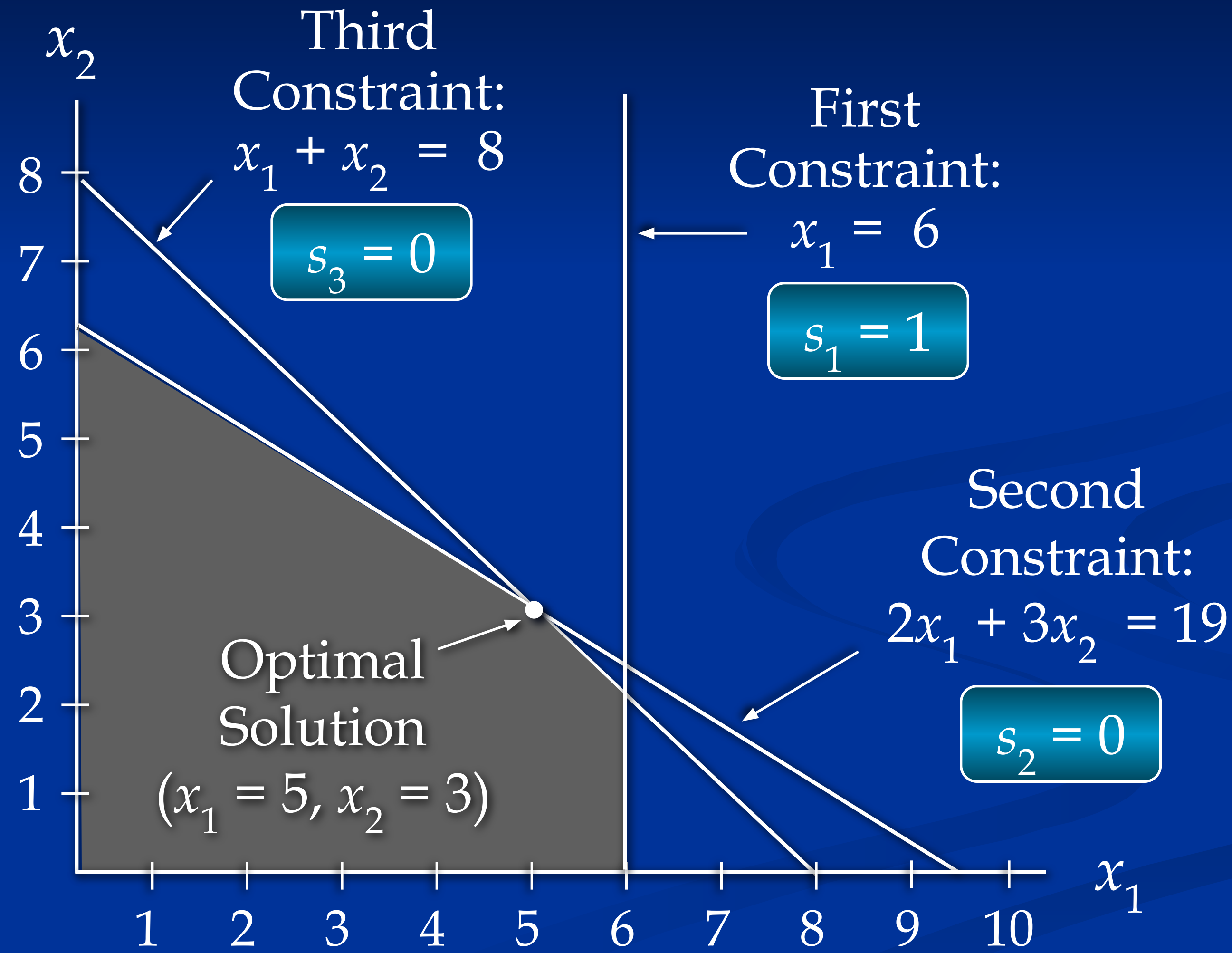
- Example 1 in Standard Form

$$
\begin{array}{ll}
\text{Max} & 5x_1 + 7x_2 + 0s_1 + 0s_2 + 0s_3 \\
\text{s.t.} & x_1 + s_1 = 6 \\
 & 2x_1 + 3x_2 + s_2 = 19 \\
 & x_1 + x_2 + s_3 = 8 \\
 & x_1, x_2, s_1, s_2, s_3 \geq 0
\end{array}
$$

s_1, s_2, and s_3 are slack variables

Slack Variables

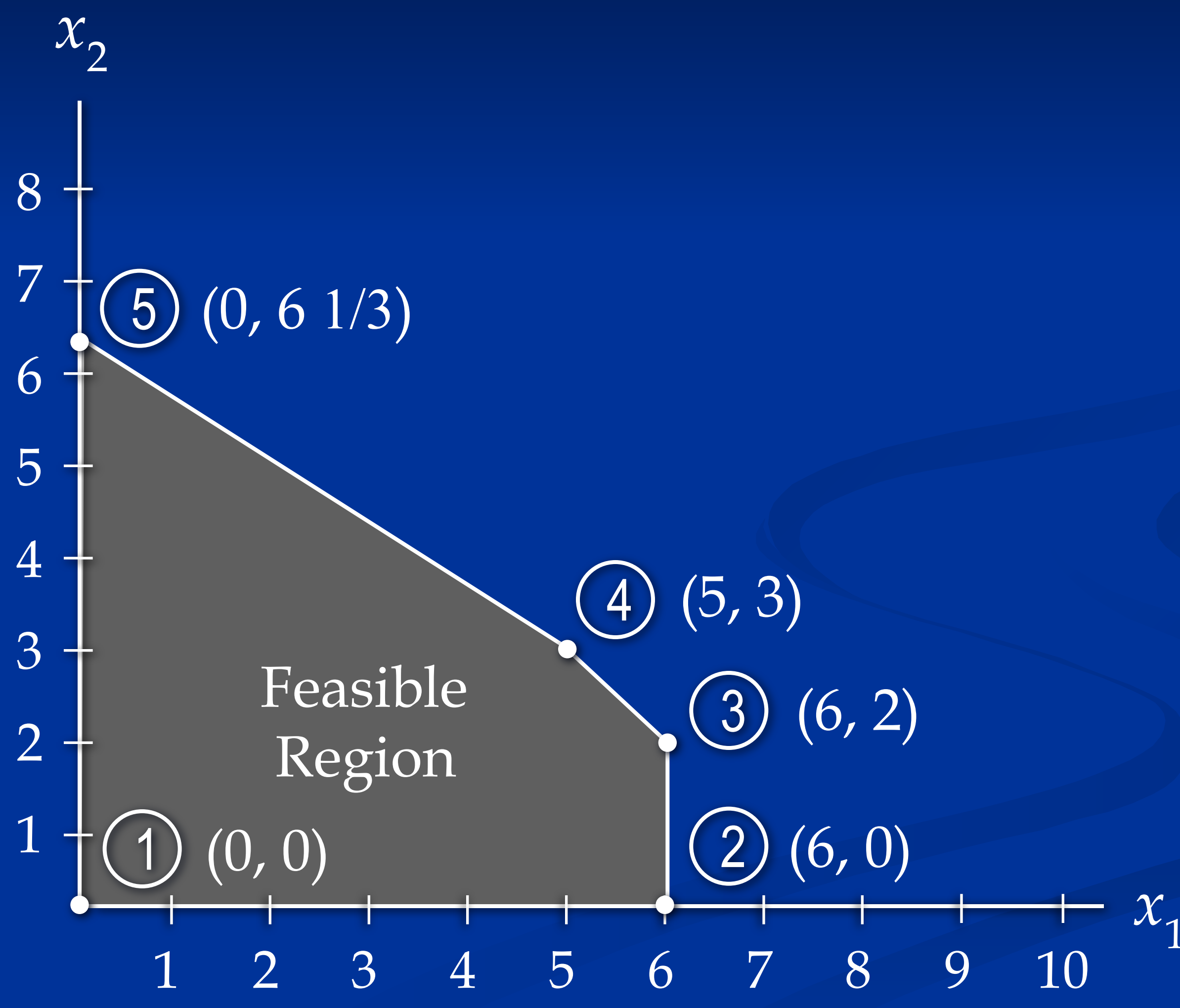
- Optimal Solution



Extreme Points and the Optimal Solution

- The corners or vertices of the feasible region are referred to as the extreme points.
- An optimal solution to an LP problem can be found at an extreme point of the feasible region.
- When looking for the optimal solution, you do not have to evaluate all feasible solution points.
- You have to consider only the extreme points of the feasible region.

Example 1: Extreme Points



Computer Solutions

- LP problems involving 1000s of variables and 1000s of constraints are now routinely solved with computer packages.
- Linear programming solvers are now part of many spreadsheet packages, such as Microsoft Excel.
- Leading commercial packages include CPLEX, LINGO, MOSEK, Xpress-MP, and Premium Solver for Excel.
- The Management Scientist, a package developed by the authors of your textbook, has an LP module.

Interpretation of Computer Output

- In this chapter we will discuss the following output:
  - objective function value
  - values of the decision variables
  - reduced costs
  - slack and surplus
- In the next chapter we will discuss how an optimal solution is affected by a change in:
  - a coefficient of the objective function
  - the right-hand side value of a constraint

Example 1: Spreadsheet Solution

- Partial Spreadsheet Showing Problem Data

	A	B	C	D
1		LHS Coefficients		
2	Constraints	X1	X2	RHS Values
3	#1	1	0	6
4	#2	2	3	19
5	#3	1	1	8
6	Obj.Func.Coeff.	5	7	

Example 1: Spreadsheet Solution

- Partial Spreadsheet Showing Solution

	A	B	C	D
8		Optimal Decision Variable Values		
9		X1	X2	
10		5.0	3.0	
11				
12	Maximized Objective Function		46.0	
13				
14	Constraints	Amount Used		RHS Limits
15	#1	5	<=	6
16	#2	19	<=	19
17	#3	8	<=	8

Example 1: Spreadsheet Solution

- Interpretation of Computer Output

We see from the previous slide that:

Objective Function Value	=	46
Decision Variable #1 (x_1)	=	5
Decision Variable #2 (x_2)	=	3
Slack in Constraint #1	=	6 – 5 = 1
Slack in Constraint #2	=	19 – 19 = 0
Slack in Constraint #3	=	8 – 8 = 0

Example 2: A Simple Minimization Problem

- LP Formulation

$$
\begin{array}{lrcl}
\text{Min} & 5x_1 + 2x_2 & & \\
\text{s.t.} & 2x_1 + 5x_2 & \geq & 10 \\
 & 4x_1 - x_2 & \geq & 12 \\
 & x_1 + x_2 & \geq & 4 \\
 & x_1, x_2 & \geq & 0
\end{array}
$$

Example 2: Graphical Solution

- Graph the Constraints

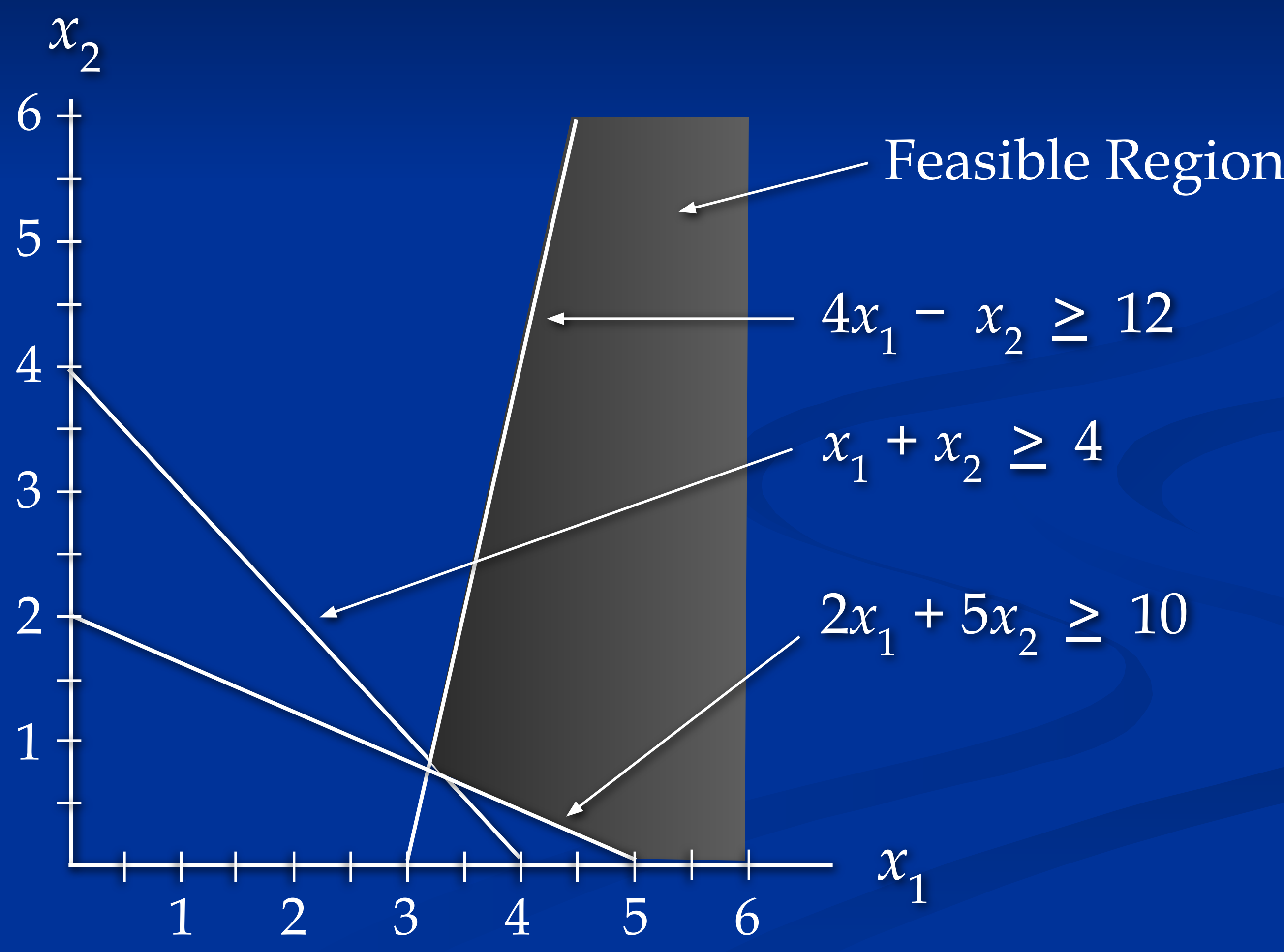
Constraint 1: When $x_1 = 0$, then $x_2 = 2$; when $x_2 = 0$, then $x_1 = 5$. Connect (5,0) and (0,2). The ">" side is above this line.

Constraint 2: When $x_2 = 0$, then $x_1 = 3$. But setting x_1 to 0 will yield $x_2 = -12$, which is not on the graph. Thus, to get a second point on this line, set x_1 to any number larger than 3 and solve for x_2: when $x_1 = 5$, then $x_2 = 8$. Connect (3,0) and (5,8). The ">" side is to the right.

Constraint 3: When $x_1 = 0$, then $x_2 = 4$; when $x_2 = 0$, then $x_1 = 4$. Connect (4,0) and (0,4). The ">" side is above this line.

Example 2: Graphical Solution

- Constraints Graphed



Example 2: Graphical Solution

- Graph the Objective Function

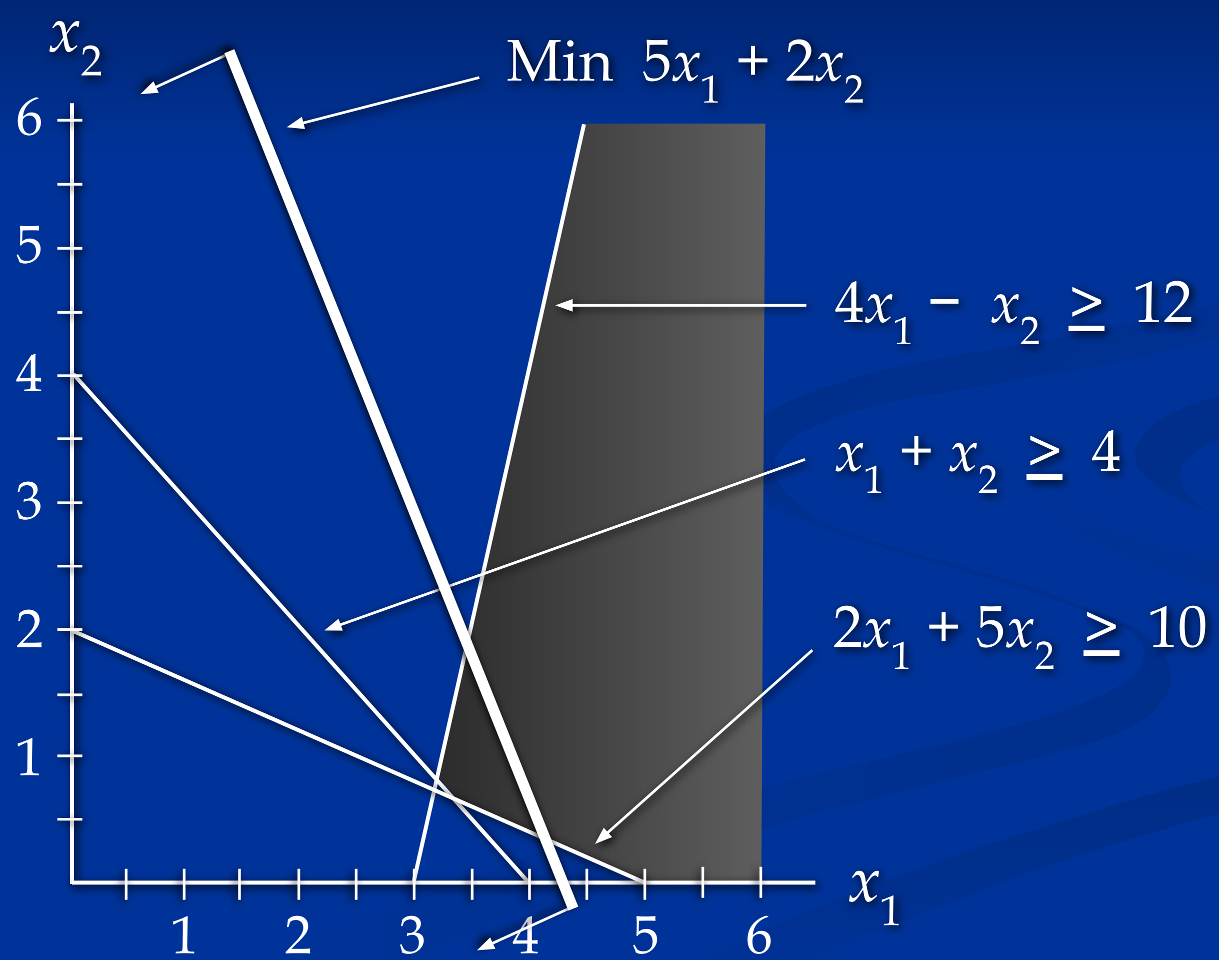
  Set the objective function equal to an arbitrary constant (say 20) and graph it. For $5x_1 + 2x_2 = 20$, when $x_1 = 0$, then $x_2 = 10$; when $x_2 = 0$, then $x_1 = 4$. Connect (4,0) and (0,10).

- Move the Objective Function Line Toward Optimality

  Move it in the direction which lowers its value (down), since we are minimizing, until it touches the last point of the feasible region, determined by the last two constraints.

Example 2: Graphical Solution

- Objective Function Graphed



Example 2: Graphical Solution

- Solve for the Extreme Point at the Intersection of the Two Binding Constraints

$$4x_1 - x_2 = 12$$
$$x_1 + x_2 = 4$$

Adding these two equations gives:

$$5x_1 = 16 \quad \text{or} \quad x_1 = 16/5$$

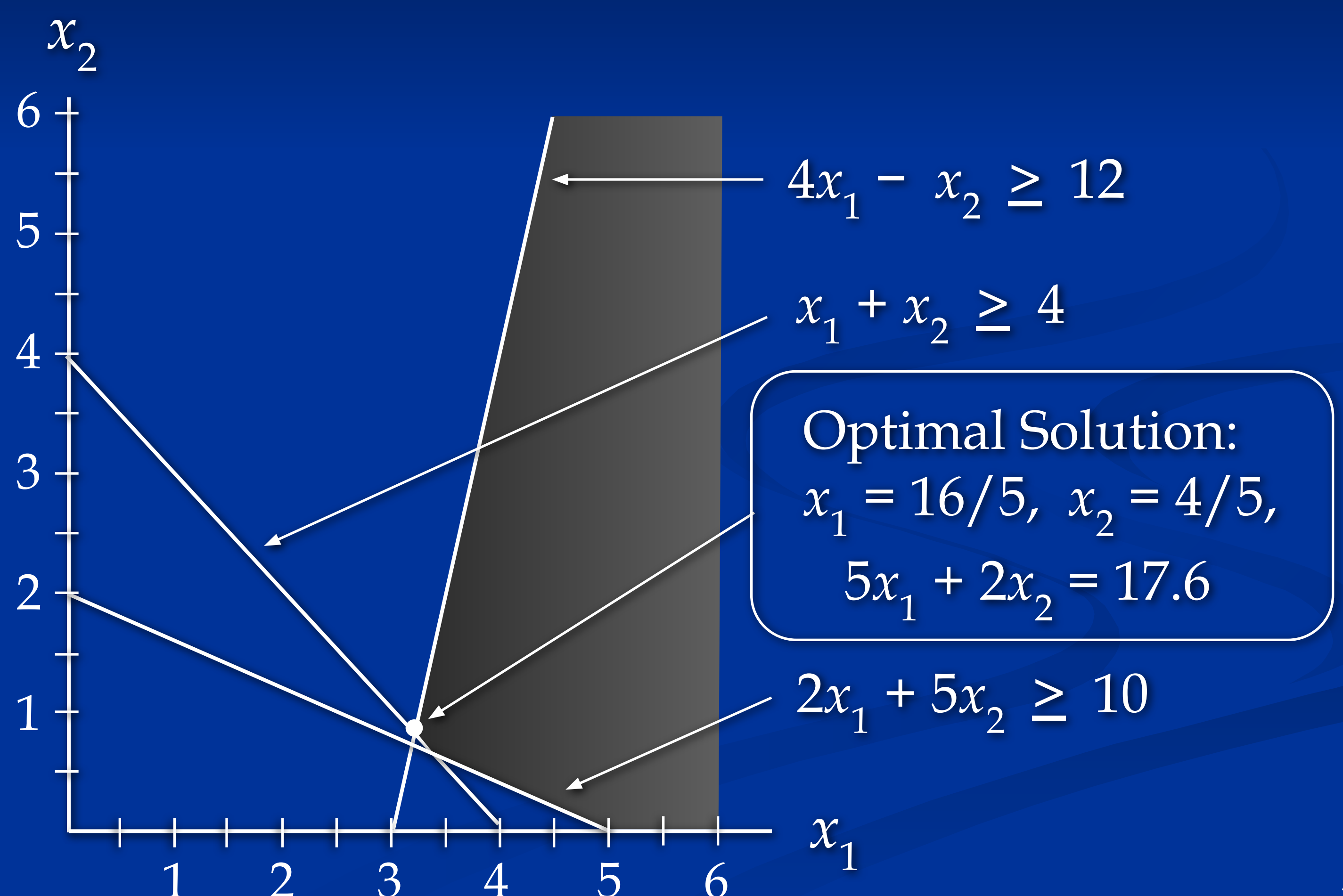
Substituting this into $x_1 + x_2 = 4$ gives: $x_2 = 4/5$

- Solve for the Optimal Value of the Objective Function

$$5x_1 + 2x_2 = 5(16/5) + 2(4/5) = 88/5$$

Example 2: Graphical Solution

- Optimal Solution



Summary of the Graphical Solution Procedure for Minimization Problems

- Prepare a graph of the feasible solutions for each of the constraints.
- Determine the feasible region that satisfies all the constraints simultaneously.
- Draw an objective function line.
- Move parallel objective function lines toward <u>smaller</u> objective function values without entirely leaving the feasible region.
- Any feasible solution on the objective function line with the <u>smallest</u> value is an optimal solution.

Surplus Variables

- Example 2 in Standard Form

$$\begin{array}{lllllll}
\text{Min} & 5x_1 + 2x_2 + 0s_1 + 0s_2 + 0s_3 & & & & & \\
\text{s.t.} & 2x_1 + 5x_2 & - s_1 & & & \geq & 10 \\
 & 4x_1 - x_2 & & - s_2 & & \geq & 12 \\
 & x_1 + x_2 & & & - s_3 & \geq & 4 \\
 & & & x_1, x_2, s_1, s_2, s_3 & & \geq & 0
\end{array}$$

s_1 , s_2 , and s_3 are surplus variables

Example 2: Spreadsheet Solution

- Partial Spreadsheet Showing Problem Data

	A	B	C	D
1		LHS Coefficients		
2	Constraints	X1	X2	RHS
3	#1	2	5	10
4	#2	4	-1	12
5	#3	1	1	4
6	Obj.Func.Coeff.	5	2	

Example 2: Spreadsheet Solution

- Partial Spreadsheet Showing Formulas

	A	B	C	D
9		Decision Variables		
10		X1	X2	
11	Dec.Var.Values			
12				
13	Minimized Objective Function		=B6*B11+C6*C11	
14				
15	Constraints	Amount Used		Amount Avail.
16	#1	=B3*B11+C3*C11	>=	=D3
17	#2	=B4*B11+C4*C11	>=	=D4
18	#3	=B5*B11+C5*C11	>=	=D5

Example 2: Spreadsheet Solution

- Partial Spreadsheet Showing Solution

	A	B	C	D
9		Decision Variables		
10		X1	X2	
11	Dec.Var.Values	3.20	0.800	
12				
13	Minimized Objective Function		17.600	
14				
15	Constraints	Amount Used		Amount Avail.
16	#1	10.4	>=	10
17	#2	12	>=	12
18	#3	4	>=	4

Example 2: Spreadsheet Solution

- Interpretation of Computer Output

We see from the previous slide that:

Objective Function Value	=	17.6
Decision Variable #1 (x_1)	=	3.2
Decision Variable #2 (x_2)	=	0.8
Surplus in Constraint #1	=	10.4 − 10 = 0.4
Surplus in Constraint #2	=	12.0 − 12 = 0.0
Surplus in Constraint #3	=	4.0 − 4 = 0.0

Special Cases

- Infeasibility
  - No solution to the LP problem satisfies all the constraints, including the non-negativity conditions.
  - Graphically, this means a feasible region does not exist.
  - Causes include:
    - A formulation error has been made.
    - Management's expectations are too high.
    - Too many restrictions have been placed on the problem (i.e. the problem is over-constrained).

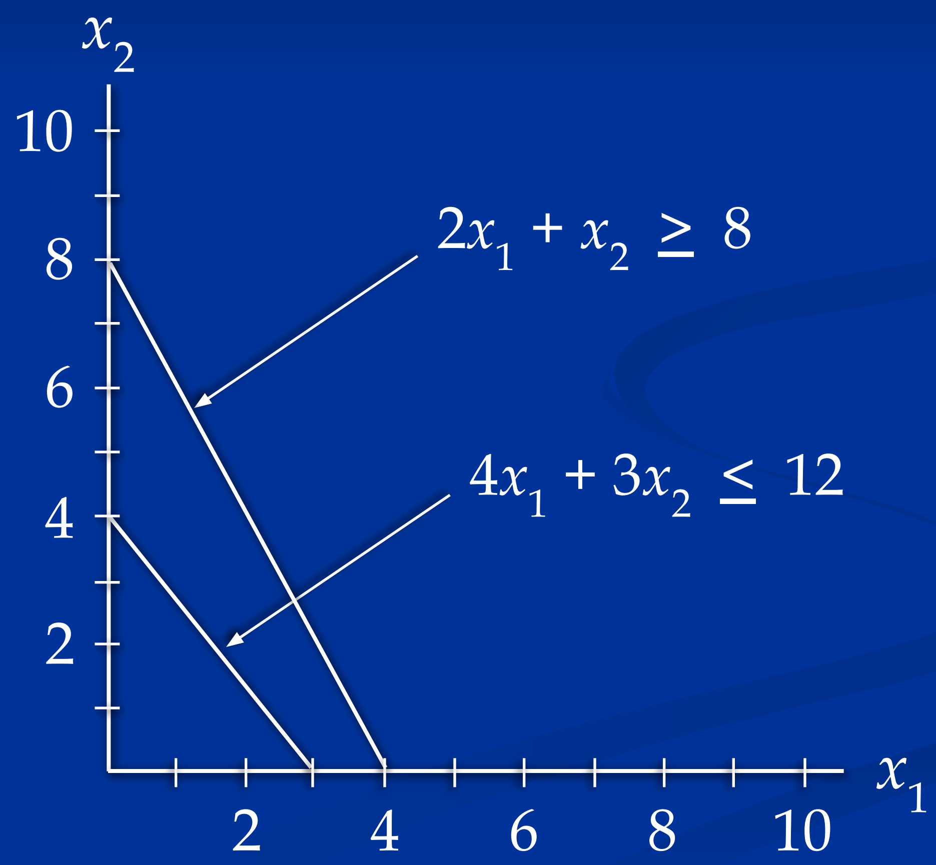
Example: Infeasible Problem

- Consider the following LP problem.

$$
\begin{array}{ll}
\text{Max} & 2x_1 + 6x_2 \\
\\
\text{s.t.} & 4x_1 + 3x_2 \le 12 \\
 & 2x_1 + x_2 \ge 8 \\
\\
 & x_1, x_2 \ge 0
\end{array}
$$

Example: Infeasible Problem

- There are no points that satisfy both constraints, so there is no feasible region (and no feasible solution).



Special Cases

- Unbounded
  - The solution to a maximization LP problem is unbounded if the value of the solution may be made indefinitely large without violating any of the constraints.
  - For real problems, this is the result of improper formulation. (Quite likely, a constraint has been inadvertently omitted.)

Example: Unbounded Solution

- Consider the following LP problem.

$$
\begin{aligned}
\text{Max} \quad & 4x_1 + 5x_2 \\
\text{s.t.} \quad & x_1 + x_2 \geq 5 \\
& 3x_1 + x_2 \geq 8 \\
& x_1, x_2 \geq 0
\end{aligned}
$$

Example: Unbounded Solution

- The feasible region is unbounded and the objective function line can be moved outward from the origin without bound, infinitely increasing the objective function.